

Member 4 — Final Work Report (Phases 1–7)

Project: Traffic Safety Intelligence — Near-Miss Detection, Accident Detection, and Safety Risk Fusion **Member role:** Member 4 — Safety Intelligence / Model Layer **Report date:** 2026-09-03 **Status:** Phases 1–7 complete · Phase 8 (this report) · Members 5–10 phases pending **Verification:** every metric in this report was read from the project’s own reports, JSON summaries, and model metadata on 2026-09-03, and the full test suite was re-run on that date (**119 passed, 0 failed**).

Table of Contents

- 1. [Member 4 Role](#)
- 2. [Datasets](#)
- 3. [Phase-by-Phase Work](#)
- 4. [Near-Miss System](#)
- 5. [Accident Detection System](#)
- 6. [Safety Risk Fusion](#)
- 7. [Final Member 4 Output](#)
- 8. [What Is Actually Working](#)
- 9. [Testing](#)
- 10. [File / Artifact Inventory](#)
- 11. [Limitations](#)
- 12. [Member 5 Handoff](#)
- 13. [Future Work](#)
- 14. [Final Summary](#)

1. Member 4 Role

1.1 Purpose

Member 4 owns the **safety-intelligence modeling layer** of the project: the path from raw traffic data to a fused, explainable safety-risk assessment. Member 4 takes the two raw datasets (a DTC-FM traffic-conflict feature table and an accident CCTV object-detection corpus), audits and prepares them, trains and evaluates one probability model per dataset, calibrates both models’ outputs into probabilities, and fuses the probabilities with caller-supplied severity and recency into a single transparent risk score with a LOW/MEDIUM/HIGH/CRITICAL level.

1.2 Responsibilities (as executed in Phases 1–7)

Responsibility	Phases
Dataset audit, quality grading, leakage screening, dataset identification	1
Data preparation (processing artifacts, feature policy, split-integrity verification)	2
Near-miss train/validation/test split + model training + model selection	3
Near-miss probability calibration, error/robustness analysis, probability contract	4
Accident detector: baseline training, failure diagnosis, fix + improvement experiments	5, 5B
Accident detector: checkpoint selection, threshold study, probability calibration, single frozen test evaluation	6
Risk fusion engine + orchestrator + config, integration of both finished models	7

1.3 What Member 4 does NOT control

- **Traffic-signal control / actuation.** No Member 4 output may be used to control signals. Every artifact explicitly documents this (`risk_engine.py`, `probability_contract.md`, phase 6/7 reports). Signal-control and decision-making belong to later phases / Member 5.
- **Severity measurement.** Neither dataset carries severity labels. `severity` is a caller-supplied number in [0, 1]; Member 4 does not measure or define it.
- **Event timestamps.** `event_age_hours` is caller-supplied; Member 4 only applies the configured decay.
- **Label-semantics authority.** The DTC-FM class-1 semantics are unconfirmed; Member 4 deliberately does not hard-code an interpretation and reports this as an open item.
- **Dataset modification.** Original files under `dataset/` are read-only throughout; SHA-256 hashes are recorded and verified in Phase 6 metadata.
- **Accuracy optimization beyond what was done.** Detector accuracy work was explicitly deferred (Phase 6 rule: "no retraining, no architecture changes, no accuracy work").

2. Datasets

2.1 DTC-FM Near-Miss Dataset

Property	Value (verified, Phase 1 audit)
Files	<code>dataset/filtered_features_DTC_FM.csv</code> (2,508 × 15), <code>dataset/target_labels_DTC_FM.csv</code> (2,508 × 1)

Property	Value (verified, Phase 1 audit)
Samples	2,508 interactions
Features	13 numeric + 2 categorical (<code>class_object_1/2</code> ∈ {Car, VRU, Ped, LCV, HCV}), plus 4 engineered <code>FE_*</code> columns
Missing / infinite values	0 / 0
Duplicate rows	5 groups (10 rows, 0.20%), identical features and identical labels; 0 conflicting labels
Alignment	Positional only — the labels file has no ID column
Label column	<code>event_type</code> ∈ {0.0, 1.0}
Class distribution	<code>class_0</code> : 1,867 (74.44%) · <code>class_1</code> : 641 (25.56%) — moderately imbalanced (~2.9:1)
Quality grade	GOOD

Confirmed/unconfirmed semantics. The files contain **no label map**. Class-conditional statistics (`class_1` rows have shorter PET — 1.85 s vs 2.38 s, smaller minimum distance — 2.04 m vs 6.26 m, smaller pixel distance — 26 px vs 104 px, larger encounter angle — 47.8° vs 29.5°) are *consistent with* "1 = conflict/near-miss", but this was never documentation-confirmed. All downstream artifacts therefore use neutral `class_0` / `class_1` terminology and carry an explicit unconfirmed-semantics caveat. This remains the single most important open data item for the project.

Engineered features (reverse-engineered with evidence):

Feature	Verified formula	Max abs error	Status
<code>FE_inv_PET</code>	<code>1 / PET</code>	3.6e-06	redundant transform
<code>FE_dist_squared</code>	<code>target_dist_px²</code>	5.8e-11	redundant transform
<code>FE_log_mesafe</code>	<code>ln(1 + target_dist_px)</code>	8.9e-16	redundant transform
<code>FE_safety_index</code>	<code>(speed1 + speed2) / (2 · max(target_dist_px, 1e-6))</code> — resolved in Phase 3	0.0012	redundant, numerically unstable; excluded from the primary config

Leakage screening (Phase 1 → 3): no ID columns exist (no ID-based leakage possible); no target-derived features exist; `FE_safety_index` was downgraded MEDIUM → NONE once its formula was resolved (it is a pure transform of three *input* features, and including it never improved validation metrics); `PET` is a **post-event** measurement — a deployment-availability design concern, not statistical leakage; pixel-distance features are camera-geometry-dependent (domain-shift risk).

2.2 Accident CCTV Dataset

Property	Value (verified, Phase 1 audit)
Source	Hugging Face <code>justjuu/traffic-accident-cctv-object-detection</code> (CC0-1.0; Roboflow "Accident and Non-accident" CCTV, letterboxed 640×640, train-only flip/brightness augmentation)
Format	4 Parquet shards, HF image-object-detection schema: <code>image{bytes: PNG, path: null} + objects{bbox[], category[]}</code>
Images	2,763 total, all 640×640 PNG (2,122 train / 316 validation / 325 test)
Objects	5,400 COCO-style <code>[x, y, w, h]</code> absolute-pixel boxes (4,120 / 678 / 602 per split); 1–12 objects per image (avg 1.95)
Classes	<code>0 = accident</code> , <code>1 = non_accident</code> — confirmed from the source dataset card (not embedded in the files); recorded in <code>data/processed/accident/label_mapping.json</code>
Box validity	100% of boxes within 640×640 bounds; <code>bbox</code> format verified empirically as <code>xywh</code> (not <code>corners</code>)
Duplicates	26 exact-byte duplicate images inside train only (1.2%); 0 cross-split duplicates (SHA-256 over all 2,763 PNGs)
Corruption	0 (IHDR validated on all images, decode-sampled every 100th)
Schema	byte-identical Arrow schema across all 4 shards
Quality grade	GOOD

Object-level class distribution:

Split	accident	non_accident	accident %
train	2,313	1,807	56.14%
validation	338	340	49.85%
test	391	211	64.95%

Image-level prevalence: 98.59% (train) / 99.37% (validation) / 99.69% (test) of frames contain ≥1 accident object. Nearly every frame is accident-positive — this is why the accident task is framed as **object-level detection/localization**, and it later weakly identifies the event-level calibration (\$5, \$11). A train→test object-level accident-rate drift of +8.8 points is documented.

2.3 Dataset Splits

Dataset	Split strategy	train	validation	test
DTC-FM near-miss	group-aware, deficit-based greedy stratified, seed 42, duplicate groups kept whole	1,756 (70.02%)	376 (14.99%)	376 (14.99%)
Accident CCTV	pre-existing provider split, verified content-clean	2,122	316	325

Near-miss split class ratios are preserved per split (74.4/25.6); integrity checks: 0 duplicate groups crossing splits, 0 identical feature rows crossing splits, 0 `sample_index` overlap, every sample assigned exactly once (1,756 + 376 + 376 = 2,508). The split is reproducible (`src/near_miss/split.py` , fixed seed, no library split shortcut).

2.4 Data-Quality and Leakage Checks (summary)

Check	Near-miss (DTC-FM)	Accident CCTV
Missing values	0	0 (no empty PNG bytes, no empty annotations)
Duplicates	5 groups / 10 rows, benign, kept	26 train-internal exact duplicates, kept, documented
Cross-split contamination	0 (group-aware split)	0 (SHA-256 verified)
Label-map availability	UNCONFIRMED (neutral class_0/class_1)	Confirmed via source dataset card
Leakage found	none statistical; PET post-event design concern; FE_safety_index resolved to a benign transform	none (no IDs/timestamps exist)
Split integrity	PASS	PASS (<code>split_integrity_report.json</code>)
Schema consistency	n/a (single CSV pair)	identical across 4 shards

3. Phase-by-Phase Work

Phase 1 — Dataset Audit

Item	Content
Objective	Read-only audit of all six source files: identification, quality grading, duplicates, leakage screening, risk register for later phases.
Work performed	File-by-file inventory; CSV deep audit; Parquet metadata/streaming audit (no full load); bbox format probe; PNG integrity checks; SHA-256 dedup across all 2,763 images; engineered-feature reverse-engineering (evidence-based).
Key decisions	Treat the two dataset families as separate modules (never combined); classify the Parquet corpus as "D. Accident + object detection"; flag DTC-FM label semantics as unconfirmed and FE_safety_index as MEDIUM-suspicious; originals untouched.
Methods	<code>audit_scripts/csv_audit.py</code> , <code>parquet_audit.py</code> , <code>parquet_labels_audit.py</code> , <code>bbox_format_probe.py</code> ; SHA-256 hashing; IHDR parsing; statistical class-conditional profiling.
Outputs	<code>reports/dataset_inventory.md</code> , <code>near_miss_dataset_audit.md</code> , <code>accident_dataset_audit.md</code> , <code>data_quality_report.md</code> , <code>dataset_audit.md</code> , <code>phase1_summary.json</code> .
Tests	None (audit-only phase; all checks are in-script verifications).
Results	All datasets graded GOOD ; 0 missing/corrupt; 0 cross-split duplicates; risk register of 5 items for later phases; no models trained.
Limitations	DTC-FM label semantics unconfirmed; FE_safety_index formula unidentified at this point; no ground truth for "quality" beyond internal consistency checks.

Phase 2 — Data Preparation

Item	Content
Objective	Produce processed, verified, model-ready artifacts without modifying originals.
Work performed	Positional DTC-FM merge (no sorting, no pre-merge drops) into <code>dtc_fm_combined.csv</code> + exact-float <code>.parquet</code> companion; per-feature keep/flag policy; accident index artifacts (<code>train/validation/test_index.parquet</code>) referencing (<code>source_file</code> , <code>row_index</code>) without copying image bytes; duplicate reports; split-integrity report; corruption report; data dictionary.
Key decisions	Preserve all rows (dedup deferred); nothing deleted, no scaler fitted, no feature selection; PET kept with a post-event flag ; FE_safety_index flagged suspicious pending Phase 3; label mapping recorded with explicit unconfirmed status for DTC-FM and confirmed mapping for accident; byte-exact Parquet companion created because CSV round-trip shifts the last ULP of 23 floats (max 9e-16).
Outputs	<code>data/processed/near_miss/</code> (combined CSV/Parquet, duplicate report, label mapping, feature policy), <code>data/processed/accident/</code> (3 index parquets, duplicate report, split-integrity report, label mapping, corruption report), <code>reports/phase2_data_preparation.md</code> , <code>data_dictionary.md</code> , <code>class_balance_phase2.json</code> .
Tests	Post-preparation validation: features byte-identical in Parquet artifact; labels unchanged; index counts match scans (2,122/316/325); zero cross-split contamination; zero writes under <code>dataset/</code> — all PASS .
Results	2,508 + 2,763 records processed with zero loss; 6 documented data-quality issues carried forward.
Limitations	DTC-FM alignment is positional (any filtering must preserve index alignment); camera-dependent pixel features noted; splitting deferred to Phase 3.

Phase 3 — Near-Miss Split + Model Training

Item	Content
Objective	Create a leakage-safe split and train probability-producing near-miss/conflict classifiers.
Work performed	Group-aware stratified 70/15/15 split (seed 42); trained 10 scikit-learn pipelines (Dummy baseline + {Logistic Regression, Random Forest, Gradient Boosting} × {A, B, D}); full threshold analysis; feature importances; leakage review.
Key decisions	Configs: A = offline full minus FE_safety_index (14 features), B = A + FE_safety_index (15), D = deployment-realistic, PET-family excluded (12). Preprocessing (median imputation, standard scaling, one-hot with <code>handle_unknown="ignore"</code>) fitted on train only inside each saved pipeline. Class imbalance handled via class weights, not resampling. Selection on validation metrics only — no test metric entered model selection.
Models	DummyClassifier (majority), LogisticRegression, RandomForestClassifier, GradientBoostingClassifier (all exposing <code>predict_proba()</code>).
Outputs	<code>models/near_miss/{run_name}.joblib</code> (10 pipelines), splits + <code>split_assignments.csv</code> , <code>model_comparison.csv</code> , <code>model_results.json</code> , <code>threshold_analysis.csv</code> , <code>feature_importance.csv</code> , <code>split_report.md</code> , <code>leakage_review.md</code> , 5 figures.
Tests	<code>tests/test_near_miss_phase3.py</code> — 7 tests (split integrity, feature-list integrity, <code>sample_index</code> exclusion, pipeline contract).

Item	Content
Results	Test ROC-AUC 0.89–0.93 across real models vs 0.500 dummy; RF-A best (test ROC-AUC 0.929, PR-AUC 0.830); leakage review PASS .
Limitations	Label semantics unconfirmed (neutral class_0/class_1 metrics); RF memorizes train (train PR 0.999) → calibration deferred to Phase 4; models are offline-only while PET is included.

Phase 4 — Near-Miss Evaluation + Calibration

Item	Content
Objective	Calibrate the selected model's probabilities, analyze errors and robustness, define the probability contract.
Work performed	Verified split integrity; compared raw/sigmoid/isotonic calibration fitted on validation (n=376); reliability diagrams; calibrated threshold analysis; FN/FP error analysis; validation→test robustness deltas; probability contract.
Key decisions	Sigmoid (Platt) selected — isotonic scored lower in-sample but showed MCE ≈ 0 on validation (overfit signature on n=376); sigmoid chosen for stability. Raw probabilities exposed as ranking scores; the risk engine (Phase 7) consumes calibrated probabilities, not hard thresholds.
Outputs	models/near_miss/random_forest_A_calibrated.joblib (+ meta JSON), calibration_report.md, calibration_comparison.csv, threshold_analysis_calibrated.csv, error_analysis.md, robustness_analysis.md, probability_contract.md, 4 figures.
Tests	tests/test_near_miss_phase4.py — 18 collected tests (calibration metrics, artifact round-trip, contract).
Results	See §4, Test Brier 0.0998 → 0.0898; test ECE 0.086 → 0.044 (sigmoid).
Limitations	Calibration fitted on a small validation set (376 samples); calibrated recall at t=0.5 drops to 0.6875 (recall-first operating points available at lower thresholds); label semantics still unconfirmed.

Phase 5 — Accident Detector Baseline

Item	Content
Objective	Train and evaluate a first accident object detector.
Work performed	Environment setup (.venv-accident , Python 3.12.10, torch 2.14.0+cpu, CPU-only, 8 threads); Faster R-CNN training; COCO-style evaluation; debug visualizations.
Key decisions	fasterrcnn_mobilenet_v3_large_320_fpn (COCO-pretrained, trainable_backbone_layers=3), 320×320 inputs (CPU practicality), batch 4, SGD lr 5e-3, 2 epochs, seed 42.
Outputs	models/accident/{best,final}_detector.pth, model_metadata.json, training_history.json, training_config.json, validation/test results reports, environment_report.md, accident_detection_contract.md, debug + ground-truth figures.
Tests	tests/test_accident_phase5.py — 19 tests (record loading, box geometry, loss finiteness, tensor shapes, inference contract).
Results	Baseline was effectively broken: validation mAP@50 0.0948 , accident AP@50 0.0000 ; test mAP@50 0.0987. See Phase 5B for the root cause.
Limitations	Misconfigured label mapping (diagnosed in 5B); 2 epochs is far from convergence; aggregate-only loss logging hid the problem; tests verified code mechanics, not learning correctness.

Phase 5B — Accident Detector Diagnosis + Improvement

Item	Content
Objective	Root-cause the Phase 5 failure and fix it, keeping the baseline checkpoint immutable.
Work performed	Full-path box-pipeline audit; empirical prediction probe over the entire validation split; visual debugging of 24 validation images; label-space fix; two tracked experiments.
Key decision / root cause	Confirmed label-mapping bug (primary cause): torchvision detectors reserve internal label 0 for background. Phase 5 passed dataset labels (0=accident, 1=non_accident) un-shifted, so every accident annotation was supervised as background and the only foreground class the model ever saw was non_accident → accident AP = 0.0 by construction. Probe evidence: predicted model labels {1: 981} (label 0 never emitted), mean emitted confidence 0.156. Fix: src/accident/labels.py single source of truth with a +1 offset (dataset 0/1 → model 1/2, model 0 = background), NUM_CLASSES = 3, per-component loss logging, background-masked inference mapping labels back to dataset space.
Experiments	EXP_A — mapping fix only, 3 epochs (1,306 s): val mAP@50 0.4686 , mAP@50:95 0.2313 , accident AP@50 0.7624 , accident recall 0.5083. EXP_B — fixed mapping, 25 epochs/early-stop 6/StepLR@8; interrupted after epoch 1 (val accident AP 0.7114, mAP@50 0.4047 — inferior to EXP_A; interrupted state retained for future accuracy work). EXP_C — not run: pretrained-backbone use was already verified in code/test. EXP_D — reserved, not needed.
Outputs	src/accident/labels.py (+ model/train/inference updates), phase5b_expA_best.pth, phase5b_expB_best*.pth, phase5b_interrupted_checkpoint.pth, phase5b_checkpoint_metadata.json, phase5b_experiment_log.md, phase5b_experiments.csv, histories + run logs, threshold-sweep table.
Tests	tests/test_accident_phase5b.py — 16 tests (mapping round-trip, no background labels in training targets, 3-class head width, full-dataset box validity, save/load, checkpoint selection, threshold analysis); all 19 Phase 5 tests still pass.
Results	The background/label collision alone explains the Phase 5 accident AP of 0; after the +1 offset, accident AP@50 = 0.76 after only 3 epochs with all other hyperparameters unchanged.
Limitations	320×320 resolution kept (640 would be ~4× slower/epoch on this CPU); EXP_B incomplete; confidence values remain uncalibrated detector scores.

Phase 6 — Accident Evaluation + Confidence Calibration

Item	Content
Objective	Select the best checkpoint on validation, derive and calibrate an event-level accident probability, run exactly ONE frozen test evaluation. No retraining, no architecture changes, no accuracy work (deferred).
Work performed	Checkpoint selection (validation-only, rule: max validation accident AP@50, mAP@50 tie-break); confidence threshold study; three candidate aggregation signals compared; Platt vs isotonic calibration with 5-fold CV tie-breaking; reliability tables; single frozen test run; SHA-256 verification that original Parquet files were unchanged.
Key decisions	Checkpoint: phase5b_expA_best.pth (val accident AP@50 0.7624 vs EXP_B 0.7114 vs baseline 0.0). Signal: noisy_or_top5 (uncalibrated validation Brier 0.0427 vs max 0.0743, top3_mean 0.2084). Method: Platt (Brier gap vs isotonic 0.0002 = near tie; Platt CV log-loss 0.0382 vs isotonic 0.1058 decided it). Calibration fitted on validation

Item	Content
	only ; test touched once, after freezing.
Outputs	models/accident/calibrated_accident_detector.pth (detector weights + calibrator), calibration_metadata.json (incl. dataset SHA-256 hashes), calibration_results.md, threshold_analysis.csv, test_results_phase6.md/.json, phase6_summary.md/.json, probability_contract.md, 3 figures.
Tests	tests/test_accident_phase6.py — 23 tests. Suite after Phase 6: 83 passed, 0 failed (all 60 prior tests still green).
Results	See §5.
Limitations	Event-level calibration is weakly identified at ~99% accident prevalence; 5 of 324 accident-positive test images received no accident detection above the 0.05 floor; the single accident-negative test image scored 0.994 (confident event-level FP).

Phase 7 — Safety Risk Fusion

Item	Content
Objective	Fuse the two finished probability models with caller-supplied severity and recency into a transparent, configurable risk score and level. Consume both models as-is — no retraining, no artifact modification.
Work performed	Built src/config.py (all fusion parameters), src/risk_engine.py (pure fusion + validation), src/safety_intelligence.py (lazy-loading orchestrator), 36 tests, this summary report.
Key decisions	Linear weighted fusion with renormalization over present inputs; severity and recency are caller-supplied with an explicit [0, 1] contract; raw detector confidence is <i>never</i> accepted as a probability (carried separately with a not-a-probability note; refusal if the calibrated field is absent); no weights/thresholds tuned on any split; the engine raises rather than guessing on invalid or all-missing inputs.
Outputs	src/config.py, src/risk_engine.py, src/safety_intelligence.py, tests/test_risk_engine.py, reports/safety/phase7_summary.md.
Tests	36 risk-engine tests; full suite 119 passed, 0 failed .
Results	See §6.
Limitations	The risk score is an engineering heuristic — no ground truth for "risk" exists, no calibration/validation of the score itself, no signal-control path.

Phase Summary Table

Phase	Objective	Primary deliverables	Final result
1	Dataset audit	5 audit reports	All datasets GOOD; 0 cross-split contamination
2	Data preparation	Processed indexes + combined DTC-FM + policy/quality artifacts	Zero data loss; all validation checks PASS
3	Near-miss split + models	10 pipelines, split, comparison reports	RF-A selected: test ROC-AUC 0.929
4	Calibration + evaluation	Calibrated RF artifact + contracts	Test Brier 0.0898, ECE 0.044 (sigmoid)
5	Accident baseline	Faster R-CNN checkpoint + reports	Broken baseline diagnosed (accident AP 0.0)
5B	Fix + improve	Fixed 3-class training, EXP_A checkpoint	Val accident AP@50 0.7624 (from 0.0)
6	Calibration + frozen test	Calibrated detector + contracts	Test accident AP@50 0.6706; event Brier 0.0169
7	Risk fusion	Risk engine + orchestrator	Transparent fusion, 36 tests, 119 total pass

4. Near-Miss System

4.1 Preprocessing

Inside each saved scikit-learn pipeline (fitted on **train only**): median imputation → standard scaling for numeric features; one-hot encoding with `handle_unknown="ignore"` for `class_object_1/class_object_2` (HCV appears in only 2 rows — treated as rare). No scaler is fitted outside the pipeline; no feature selection beyond the documented config; `sample_index`, `event_type`, and `split` are stripped before modeling (verified by tests).

4.2 Features

Three configurations (all documented in `feature_policy.json` and `model_comparison.csv`):

Config	n	Contents
A (primary, offline)	14	9 base numeric + 2 categorical + <code>FE_inv_PET</code> , <code>FE_log_mesafe</code> , <code>FE_dist_squared</code> (excludes <code>FE_safety_index</code>)
B	15	A + <code>FE_safety_index</code>
D (deployment-realistic)	12	A minus <code>PET</code> and <code>FE_inv_PET</code>

`FE_safety_index` is excluded from the primary configuration on parsimony and numerical-stability grounds (the 1e-6 distance floor produces quantized extremes up to 2.85e7); empirically it never improved any validation metric.

4.3 Split Strategy

Group-aware, deficit-based greedy stratified assignment, seed 42: duplicate feature-row groups (5 groups × 2 rows) are kept whole within one split; classes stratified to 74.4/25.6 per split; realized 70.02/14.99/14.99. Integrity: 0 crossing groups, 0 identical rows across splits, 0 `sample_index` overlap — all verified and re-verified in Phase 4.

4.4 Models Tested

10 pipelines (all seed 42): DummyClassifier (majority) + {Logistic Regression, Random Forest, Gradient Boosting} × {A, B, D}. Full table: `reports/near_miss/model_comparison.csv`.

4.5 Selected Model

`random_forest_A` (config A, 14 features) → `random_forest_A_calibrated` (sigmoid, Phase 4). Selection used **validation metrics only**:

- 1. Best validation ROC-AUC (0.9305) and PR-AUC (0.8540) among all real models.
- 2. Best validation F1 for the minority class (0.7817).
- 3. Not the best raw recall (LR-A 0.865) — but the threshold analysis shows RF-A reaches the same recall band with higher precision at t≈0.30.
- 4. Stable val→test generalization (ROC 0.9305 → 0.9290; PR 0.8540 → 0.8304).
- 5. Documented caveat: RF fits train almost perfectly (train PR ≈ 0.9996) — the reason Phase 4 calibration was mandatory.

4.6 Validation / Test Metrics

Phase 3 experiment table (test split, threshold 0.50, minority class = class_1):

Model	Config	Accuracy	Precision (c1)	Recall (c1)	F1 (c1)	ROC-AUC	PR-AUC	FN	FNR
Dummy (majority)	A	0.745	0.000	0.000	0.000	0.500	0.256	96	1.000
Logistic Regression	A	0.806	0.586	0.813	0.681	0.912	0.818	18	0.188
Logistic Regression	B	0.809	0.590	0.823	0.687	0.913	0.813	17	0.177
Logistic Regression	D (deploy)	0.790	0.559	0.833	0.669	0.898	0.773	16	0.167
Random Forest	A	0.867	0.713	0.802	0.755	0.929	0.830	19	0.198
Random Forest	B	0.862	0.693	0.823	0.752	0.930	0.835	17	0.177
Random Forest	D (deploy)	0.854	0.688	0.781	0.732	0.912	0.777	21	0.219
Gradient Boosting	A	0.846	0.661	0.813	0.729	0.916	0.836	18	0.188
Gradient Boosting	B	0.840	0.650	0.813	0.722	0.917	0.839	18	0.188
Gradient Boosting	D (deploy)	0.843	0.655	0.813	0.726	0.901	0.749	18	0.188

Confusion matrix — `random_forest_A`, TEST, t=0.50 (Phase 3 raw model): TN 249 · FP 31 · FN 19 · TP 77.

Validation ranking: RF-A best ROC-AUC 0.9305 / PR-AUC 0.8540 / F1 0.7817; LR-A best recall 0.8646 (FN 13). Validation→test deltas for every model are tabulated in `robustness_analysis.md`; all models generalize within ~0.01 ROC-AUC.

4.7 Threshold Analysis

Raw model (validation, `random_forest_A`):

Threshold	Precision	Recall	F1	FP	FN
0.20	0.467	0.948	0.625	104	5
0.30	0.569	0.906	0.699	66	9
0.40	0.675	0.865	0.758	40	13
0.50	0.762	0.802	0.782	24	19
0.60	0.783	0.750	0.766	20	24
0.70	0.816	0.646	0.721	14	34
0.80	0.869	0.552	0.675	8	43

Calibrated model (validation, sigmoid): (from `threshold_analysis_calibrated.csv`)

Threshold	Precision	Recall	F1	FPR	FN
0.20	0.646	0.875	0.743	0.164	12
0.30	0.736	0.8125	0.772	0.100	18
0.50	0.787	0.729	0.757	0.068	26

No deployment threshold is permanently selected. Documented operating-point references: t≈0.30 recall-first (raw: recall 0.906, FN 9); t≈0.50 balanced (raw: F1 max 0.782). The risk engine consumes raw probabilities, not a hard threshold, so these are references, not decisions.

4.8 Calibration

Calibration fitted on **validation only** (n = 376):

Method	Val Brier	Test Brier	Val log-loss	Test log-loss	Val ECE	Test ECE	Test MCE
raw	0.096016	0.099755	0.308279	0.321270	0.078816	0.086362	0.252102
sigmoid (selected)	0.088684	0.089807	0.298774	0.301506	0.046262	0.043901	0.145450
isotonic	0.080224	0.091872	0.250184	0.618604	0.000000	0.027530	0.572400

Isotonic was rejected despite lower in-sample scores: validation MCE ≈ 0 indicated overfit on the small calibration set (its test log-loss of 0.6186 confirms this).

Calibrated test classification (t = 0.50): accuracy 0.8777, precision 0.8049, recall 0.6875, F1 0.7416, ROC-AUC 0.9290 (unchanged — calibration is monotone), PR-AUC 0.8304 (unchanged), TP/TN/FP/FN = 66/264/16/30.

4.9 Error and Robustness Analysis

- **False negatives (30, calibrated, t=0.5):** associated with long PET (mean 2.18 s), moderate distances (mean min-dist 3.49 m), mid-range angles (mean 41.9°); predicted probabilities 0.046–0.499 (median 0.290). Full profile: `error_analysis.md`.
- **False positives (16):** associated with very small distances (mean min-dist 1.41 m, mean pixel distance 15.6 px) and large angles (mean 53.6°) — physically close interactions the model ranks as conflicts.
- **Robustness:** validation→test ROC-AUC deltas within ± 0.008 for all nine real models (`robustness_analysis.md`); rankings are stable.
- **Feature importance (RF-A, association not causation):** PET 0.377, `min_dist_dual_check_m` 0.244, `target_dist_px` 0.121, `FE_log_mesafe` 0.088, `angle_degrees` 0.061.

4.10 Final Probability Output

`predict_proba()[:, 1]` from `random_forest_A_calibrated.joblib` — calibrated P(class_1), contract in `probability_contract.md`:

```
{"class_0_probability": 0.12, "class_1_probability": 0.88,
"model_version": "random_forest_A_calibrated", "calibrated": true}
```

4.11 Near-Miss Limitations

1. **PET is post-event/offline.** PET (and `FE_inv_PET`) is computable only after the interaction completes. Config D quantifies the cost of removing it: ~ 0.02 ROC-AUC and ~ 0.06 – 0.08 PR-AUC lost, but still $\text{ROC} \geq 0.90$. **No model trained here may be described as real-time-deployable while PET is included.** A live system needs pre-event surrogates (projected time-to-collision, distance rate) — future work.
2. **Label semantics unconfirmed** — `class_1` is only *statistically consistent* with a conflict/near-miss.
3. **Pixel features are camera-dependent** (`target_dist_px`, `FE_dist_squared`, `FE_log_mesafe`); metric features are the safer core.
4. **RF overfit** (train PR 0.999) — addressed by calibration for probabilities, but it remains a variance caveat.
5. Test split is small for the minority class (96 class-1 samples); metric uncertainty is material.

5. Accident Detection System

5.1 Why Object Detection, Not Image Classification

Image-level classification would be trivially biased: **$\sim 99\%$ of frames** (train 98.59%, validation 99.37%, test 99.69%) contain at least one accident-labeled object. A classifier predicting "accident" for every frame would score $\sim 99\%$ accuracy while carrying no information. The meaningful task is **object-level accident localization** — which object, where (bounding box), and with what confidence. The dataset is annotated per-object (5,400 COCO-style boxes), so detection uses the supervision that actually exists.

5.2 Architecture

- **Model:** `torchvision fasterrcnn_mobilenet_v3_large_320_fpn` — Faster R-CNN with a MobileNetV3-Large-320 backbone and FPN neck.
- **Initialization:** COCO-pretrained (`FasterRCNN_MobileNet_V3_Large_320_FPN_Weights.DEFAULT`), `trainable_backbone_layers=3`; 18,871,333 trainable parameters.
- **Label spaces (after the Phase 5B fix):** dataset {0 = accident, 1 = non_accident} $\leftrightarrow$ model {0 = background, 1 = accident, 2 = non_accident}, via a +1 offset in `src/accident/labels.py` (`torchvision` reserves internal label 0 for background).
- **Training config (Phase 5 / EXP_A):** 320 $\times$ 320 input, batch 4, SGD (lr 0.005, momentum 0.9, weight decay 5e-4), StepLR, seed 42, CPU-only (torch 2.14.0+cpu, 8 threads, no CUDA).
- **Inference:** `AccidentDetector` infers head width from the checkpoint, masks background outputs, maps model labels back to dataset categories; boxes returned as `[x1, y1, x2, y2]` absolute pixels.

5.3 Phase 5 Baseline → Phase 5B Improvement

Metric (validation)	Phase 5 baseline	5B EXP_A (fix, 3 epochs)
mAP@50	0.0948	0.4686
mAP@50:95	0.0374	0.2313
accident AP@50	0.0000	0.7624
accident recall (COCO AR)	0.0000	0.5083
test mAP@50 (baseline)	0.0987	—

Root cause (verified, primary): the Phase 5 model used `NUM_CLASSES = 2` with dataset labels passed un-shifted, so every accident annotation was supervised as **background**; the only foreground class the model ever saw was `non_accident`, and accident AP was 0.0 *by construction*. Probe over the full validation split with the immutable baseline checkpoint: predicted model labels {1: 981} (label 0 never emitted), mean emitted confidence 0.156. Bounding boxes were verified correct along the entire path (a new test validates every box in every split); the model was already COCO-pretrained; the 2-epoch duration was a real but secondary limitation.

Phase 5B experiments (validation only; TEST locked):

Experiment	Change	Val mAP@50	Val accident AP@50	Status
EXP_A	label-mapping fix only, 3 epochs	0.4686	0.7624	completed (1,306 s)
EXP_B	fix + 25 epochs, early stop 6, StepLR@8	0.4047	0.7114	interrupted after epoch 1 (checkpoint retained)
EXP_C	pretrained-backbone comparison	—	—	not needed (already pretrained, verified)

Early EXP_A threshold sweep (validation, pooled classes, IoU 0.5): t=0.10 → P 0.267 / R 0.631; t=0.30 → P 0.512 / R 0.516; t=0.50 → P 0.695 / R 0.441.

5.4 Selected Checkpoint

`models/accident/phase5b_expA_best.pth` — selected in Phase 6 by max validation accident AP@50 (0.7624 vs EXP_B 0.7114 vs baseline 0.0000), mAP@50 tie-break. The Phase 5 baseline checkpoint is kept immutable; the deployed artifact is `models/accident/calibrated_accident_detector.pth` (detector weights + fitted calibrator).

5.5 Validation Metrics (Phase 6, selected checkpoint)

Metric	Value
mAP@50	0.4686
mAP@50:95	0.2313
mAP@75	0.2121
accident AP@50	0.7624
accident AP@50:95	0.3915
accident recall (COCO AR)	0.5083
non_accident AP@50	0.1749

Confidence threshold study (validation, accident class, IoU 0.5):

Threshold	Precision	Recall	FP	FN
0.1	0.4348	0.8580	377	48
0.2	0.5644	0.8432	220	53
0.3	0.6447	0.8107	151	64
0.4	0.7038	0.7663	109	79
0.5	0.7590	0.7456	80	86
0.6	0.8087	0.7130	57	97
0.7	0.8388	0.6775	44	109
0.8	0.8680	0.6420	33	121
0.9	0.9118	0.5503	18	152

Usable precision/recall trade-off sits in the 0.30–0.50 band. A deployment threshold is **not** fixed — downstream consumers pick one from this table per their operating point.

5.6 Final Test Metrics (single frozen evaluation)

Metric	Value
mAP@50	0.4381
mAP@50:95	0.2116
accident AP@50	0.6706
accident AP@50:95	0.3344
accident recall (COCO AR)	0.4409
non_accident AP@50	0.2056
accident precision @0.5	0.6949
accident recall @0.5	0.6292

5.7 Calibration Method

- **Signal:** `noisy_or_top5` — noisy-OR aggregation of the top-5 accident-class box scores (above the 0.05 detector floor). Selected on lowest uncalibrated validation Brier (0.0427 vs 0.0743 for max, 0.2084 for `top3_mean`).
- **Method:** **Platt** (a = 0.48997, b = 3.87115), fitted on validation image-level labels (316 images, 314 accident-positive). Platt vs isotonic: validation Brier 0.0063 vs 0.0061 (near tie ≤ 0.002) → decided by 5-fold CV (Platt CV log-loss 0.0382 vs isotonic 0.1058; isotonic step functions risk extreme held-out probabilities).
- **Validation fit:** Brier 0.0063, log loss 0.0356, ECE 0.0000.
- **TEST was touched exactly once**, after the checkpoint, signal, and calibrator were frozen. Nothing was tuned after the test numbers were produced.

5.8 Calibrated Accident Probability (test, event level)

Metric	Value
Images	325 (324 accident-positive)

Metric	Value
Brier score	0.0169
Log loss	0.0673
Expected calibration error (10 bins)	0.0176
Reliability	5 images in bin [0.0, 0.1) (predicted 0.052, observed 1.0 — all misses); 320 in bin [0.9, 1.0] (predicted 0.9938, observed 0.9969)

Important caveat (documented in `calibration_results.md`): because ~99% of images are accident-positive, event-level Brier/log-loss/ECE are dominated by prevalence and are close to what a well-behaved constant predictor achieves. Discrimination is better read from the object-level AP and the threshold study. **The calibrator must be re-fit on any deployment data whose accident prevalence differs materially.**

5.9 Inference Output

Contract (`accident/probability_contract.md`); the two confidence quantities are distinct:

- `accident_detection_confidence` — raw `noisy_or_top5` signal, a **ranking score, not a probability**.
- `calibrated_accident_probability` — P(image contains an accident), through the fitted Platt calibrator.

```
{
  "detector_version": "accident_calibrated_detector_v1",
  "image_size": [640, 640],
  "detections": [
    {"bbox": [x1, y1, x2, y2], "category": "accident", "confidence": 0.83}
  ],
  "accident_detection_confidence": 0.83,
  "calibrated_accident_probability": 0.71,
  "calibration": {"method": "platt", "signal": "noisy_or_top5", "fit_split": "validation"}
}
```

Per-box `confidence` values remain uncalibrated detector scores. The output does not encode severity, accident count, or localization quality.

5.10 Accident Limitations

- Modest absolute accuracy.** Test accident AP@50 0.6706, mAP@50 0.4381, accident recall (COCO AR) 0.4409 — usable but far from production grade; accuracy work was explicitly deferred.
- Event-level calibration is weakly identified** (~99% prevalence); the calibrator is prevalence-bound.
- Failure modes (test, descriptive):** 5 of 324 accident-positive images received no accident detection above the 0.05 floor (probability floor 0.052 — these dominate log loss); the single accident-negative image scored 0.994 (confident event-level false positive).
- Small validation/test sets** (316/325 images) → metric variance is material.
- 320×320 training resolution** (source 640×640) — a CPU-practicality compromise.
- Letterboxed source frames** — scale information is partially baked in.
- Distribution shift risk** — the probability contract conditions validity on deployment frames resembling the dataset distribution.

6. Safety Risk Fusion (Phase 7)

6.1 What is fused

Four inputs, deliberately kept semantically separate (different models measuring different events are never merged into one probability):

Input	Source	Calibrated	Fit split
<code>near_miss_probability</code>	Phase 4 <code>random_forest_A_calibrated</code> (sigmoid) — calibrated P(class_1)	yes	validation
<code>accident_probability</code>	Phase 6 <code>calibrated_accident_detector</code> (Platt) — <code>calibrated_accident_probability</code>	yes	validation
<code>severity</code>	caller-supplied , relative value in [0, 1]; neither dataset carries severity labels	n/a	n/a
<code>event_age_hours</code>	caller-supplied ; converted to a recency factor	n/a	n/a

The raw detector confidence (`accident_detection_confidence`) is never accepted as a probability. When a detector output is fused, it is carried separately under `detector_raw_signal` with an explicit not-a-probability note; a detector output lacking the calibrated field is **refused** rather than substituted with the raw signal.

6.2 Actual Formula (exactly as implemented in `src/risk_engine.py`)

```
recency_factor = 0.5 ** (event_age_hours / RECENTY_HALF_LIFE_HOURS)

risk_score = SCALE * Σ( weight_i × value_i   for PRESENT inputs i )
              / Σ( weight_i               for PRESENT inputs i )

risk_level = LOW      if risk_score < 25
            MEDIUM   if 25 ≤ risk_score < 50
            HIGH      if 50 ≤ risk_score < 75
            CRITICAL  if risk_score ≥ 75
```

- Thresholds are **inclusive lower bounds** of their level (score exactly 25 → MEDIUM, 50 → HIGH, 75 → CRITICAL; unit-tested).
- Missing inputs (`None`) are **excluded** and the weights of the present inputs are **renormalized** — absence of an input is treated as absence of evidence, not evidence of absence. With no inputs at all the engine raises `RiskInputError` instead of returning a confident-looking LOW.
- Invalid inputs (probabilities outside [0, 1], negative event age, booleans, non-numeric severity, NaN/inf) raise `RiskInputError` — the engine never guesses. Severity is accepted only as a number in [0, 1]; no named-severity vocabulary is invented.

6.3 Configurable Thresholds (all in `src/config.py`; nothing hardcoded in the engine)

Parameter	Key	Default
Weight — near-miss probability	<code>RISK_WEIGHTS["near_miss"]</code>	0.35
Weight — accident probability	<code>RISK_WEIGHTS["accident"]</code>	0.40
Weight — severity	<code>RISK_WEIGHTS["severity"]</code>	0.15
Weight — recency	<code>RISK_WEIGHTS["recency"]</code>	0.10
Score scale	<code>RISK_SCORE_SCALE</code>	100.0
MEDIUM threshold (lower bound)	<code>RISK_LEVEL_THRESHOLDS["medium"]</code>	25.0
HIGH threshold (lower bound)	<code>RISK_LEVEL_THRESHOLDS["high"]</code>	50.0
CRITICAL threshold (lower bound)	<code>RISK_LEVEL_THRESHOLDS["critical"]</code>	75.0
Recency half-life	<code>RECENTY_HALF_LIFE_HOURS</code>	24.0 hours

All are overridable per call via the `config / risk_config` argument (validated: weights > 0 and finite; 0 < medium < high < critical; scale > 0; half-life > 0). **No weight or threshold was tuned on any dataset split** — the fusion is an engineering heuristic defined by `config.py`, fitted to nothing.

6.4 Worked Example (verified against the implementation)

Inputs: `near_miss_probability = 0.18146`, `accident_probability = 0.9959`, `severity = 0.4`, `event_age_hours = 6.0` (all four present; weights total 1.0).

```
recency_factor = 0.5 ** (6.0 / 24.0) = 0.840896
risk_score = 100 × (0.35×0.18146 + 0.40×0.9959 + 0.15×0.4 + 0.10×0.840896) / 1.0
            = 100 × (0.063511 + 0.398360 + 0.060000 + 0.084090)
            = 60.5961 → HIGH
```

This is an actual recorded output of `SafetyIntelligence.assess_image(...)` on real test data (see §7).

6.5 Fusion Limitations

- Not scientifically validated** — no ground truth for "risk" exists; the score is a transparent heuristic, echoed as `is_probability: false` in every output.
- Not for traffic-signal control** — the layer produces assessments only.
- Severity and recency are caller-supplied** — the pipeline measures neither.
- Linear fusion is a design choice** for transparency/testability, not an empirically derived model.
- Missing evidence is not absence of danger** — renormalization can understate risk when the strongest signal (e.g., accident probability) is unavailable; `missing_inputs` is always reported so consumers can gate on evidence completeness.
- Inherits all near-miss and accident caveats (\$4.11, \$5.10).

7. Final Member 4 Output (handoff artifact for Member 5)

7.1 What Member 4 produces

The public API is `SafetyIntelligence` (`src/safety_intelligence.py`):

- `assess_event(...)` — fuse a near-miss feature vector and/or a Phase 6 detector output (or an explicit calibrated accident probability) with caller-supplied `severity` and `event_age_hours`.
- `assess_image(image, ...)` — run the calibrated accident detector on an image (PIL / numpy / tensor), then fuse as above.
- `to_json(output)` — serialize; outputs are plain builtins, JSON-serializable by construction.

Every assessment carries: `engine_version`, both probabilities, `risk_score` (0–100), `risk_level`, `severity`, `event_age_hours`, `present_inputs` / `missing_inputs`, `model_provenance` (artifact paths, loaded flags, `retrained: false`), an optional `detector_raw_signal` block (explicitly not-a-probability), the full detector output under `detector_output` (detections with bboxes, categories, confidences, calibration block), and a formula-transparent `explanation` list.

7.2 Example Output — ACTUAL RECORDED INFERENCE

The JSON below is **not hand-written**. It was produced on 2026-09-03 by running the real deployed artifacts on real data: the first test-split DTC-FM row (`sample_index` = first row of `data/processed/near_miss/splits/test.parquet`; features: speeds 4/8 km/h, angle 29.19°, PET 2.87 s, min-dist 4.677 m, 48 objects in frame, Car/VRU pair) and the first test-split CCTV image (`test-00000-of-00001.parquet`, row 0, 1 annotated accident object), with caller-supplied `severity` = 0.4 and `event_age_hours` = 6.0. The `explanation` list is omitted here only for brevity; it is present in the real output.

```
{
  "engine_version": "safety_risk_fusion_v1",
  "near_miss_probability": 0.18146,
  "accident_probability": 0.9959,
  "risk_score": 60.5961,
  "risk_level": "HIGH",
  "severity": 0.4,
  "event_age_hours": 6.0,
  "present_inputs": ["accident", "near_miss", "recency", "severity"],
  "missing_inputs": [],
  "model_provenance": {
    "near_miss_artifact": "models/near_miss/random_forest_A_calibrated.joblib",
    "near_miss_model_loaded": true,
    "accident_checkpoint": "models/accident/calibrated_accident_detector.pth",
    "accident_model_loaded": true,
    "retrained": false
  },
  "detector_raw_signal": {
    "accident_detection_confidence": 0.9646,
    "note": "Raw detector confidence (ranking signal aggregated from accident-class box scores). This is NOT a probability; the probability is 'accident_probab:"
  },
  "detector_output": {
    "detector_version": "accident_calibrated_detector_v1",
    "image_size": [640, 640],
    "n_detections": 1,
    "detections": [
      {"bbox": [76.3, 127.22, 191.61, 168.67], "category": "accident", "confidence": 0.9322}
    ],
    "calibration": {"method": "platt", "signal": "noisy_or_top5", "fit_split": "validation"}
  }
}
```

(The example in `reports/safety/phase7_summary.md` — near-miss 0.711654, no accident input, risk 65.5281 HIGH — is likewise a real output of the fusion engine with a near-miss-only input.)

7.3 Contract Guarantees

- Probabilities outside [0, 1], invalid types, negative ages, NaN/inf → `RiskInputError`; all inputs missing → refusal (never a confident LOW).
- Raw detector confidence is never substituted for a probability.
- The two probabilities are never merged into one; `is_probability: false` marks the fused score.
- Deterministic inference for fixed inputs (eval mode, CPU); original datasets are never modified.

8. What Is Actually Working

Implemented and tested

Capability	Evidence
Read-only dataset audit with quality grades, duplicate/leakage screening	Phase 1 reports; SHA-256 checks
Deterministic data preparation with zero data loss and verified originals	Phase 2 report; post-write validation; dataset SHA-256 in Phase 6 metadata
Leakage-safe, reproducible 70/15/15 near-miss split	<code>split_report.md</code> ; 0 crossing groups (tested)
10 trained near-miss pipelines with saved preprocessing	<code>models/near_miss/*.joblib</code> ; Phase 3 tests
Calibrated near-miss probability model (sigmoid, val-fit)	<code>random_forest_A_calibrated.joblib</code> ; test Brier 0.0898, ECE 0.0439
Accident detector with corrected label mapping (Faster R-CNN, 3-class)	<code>phase5b_expA_best.pth</code> ; val accident AP@50 0.7624
Event-level calibrated accident probability (Platt, <code>noisy_or_top5</code> , val-fit)	<code>calibrated_accident_detector.pth</code> ; test Brier 0.0169
Transparent risk fusion with validation, renormalization, explanation strings	<code>risk_engine.py</code> ; 36 tests
End-to-end orchestrator producing the Member 5 payload	<code>safety_intelligence.py</code> ; recorded output in §7.2

Capability	Evidence
Full regression suite	119 passed, 0 failed (re-run 2026-09-03)

Implemented but not fully validated

Item	Gap
Deployment-realistic near-miss variant (config D, no PET)	Trained and evaluated offline (ROC $\geq$ 0.90) but never deployed/validated live; PET replacement not built
Accident detector accuracy	Usable but modest (test mAP@50 0.4381); accuracy work explicitly deferred; EXP_B long-run incomplete
Event-level accident calibration	Weakly identified at ~99% prevalence; validity conditioned on distribution match
Risk score semantics	Heuristic fusion; no ground truth, no external validation of score quality
Severity/recency inputs	Accepted and fused per contract, but no data source supplies them — caller-owned

Planned / future work

See §13.

9. Testing

9.1 Test Totals (re-run 2026-09-03: 119 passed, 0 failed, 239.8 s)

Suite	Tests	Covers
test_near_miss_phase3.py	7	Split integrity, feature lists, <code>sample_index</code> exclusion, pipeline contract
test_near_miss_phase4.py	18	Calibration metrics, artifact round-trip, probability contract
test_accident_phase5.py	19	Record loading, box geometry, loss finiteness, tensor shapes, inference contract
test_accident_phase5b.py	16	Label-mapping round-trip, no background in training targets, 3-class head, full-dataset box validity, save/load, checkpoint selection, threshold analysis
test_accident_phase6.py	23	Signal aggregation, calibration fit, threshold study, frozen-test pipeline, dataset-hash guard
test_risk_engine.py	36	Fusion math, threshold boundaries, missing/invalid inputs, config overrides, JSON contract, integration with both real artifacts
Total	119	

9.2 Important Validation Checks

- **Leakage checks:** 0 duplicate groups crossing near-miss splits; 0 identical feature rows crossing splits; 0 `sample_index` overlap; `sample_index` verified absent from every model's feature list; no target-derived features exist; `FE_safety_index` resolved to a benign input transform (no suspicious validation jump when included).
- **Calibration checks:** near-miss sigmoid test Brier 0.0898 / ECE 0.0439 (vs raw 0.0998 / 0.0864); isotonic rejected on overfit evidence (test log-loss 0.6186). Accident Platt validation fit Brier 0.0063 / ECE 0.0; 5-fold CV used to break the Platt/isotonic tie.
- **Dataset integrity:** SHA-256 over all 2,763 accident PNGs (0 cross-split duplicates, 26 benign train-internal); split-integrity report PASS; original Parquet files hash-verified unchanged before/after the Phase 6 run (`calibration_metadata.json`); DTC-FM features byte-identical in the Parquet artifact.
- **Process guards (tested):** TEST used for calibration: false; TEST used for threshold selection: false; accident TEST evaluations in Phase 6: exactly 1; near-miss model selection used validation metrics only.
- **Semantic guards:** raw detector confidence is never accepted as a probability (tested refusal); risk engine raises on all-missing inputs and on invalid ranges/types (tested); `is_probability: false` on risk output.

10. File / Artifact Inventory

models/

File	Phase	Purpose
<code>models/near_miss/random_forest_A_calibrated.joblib</code> (+ <code>..._meta.json</code>)	4	Deployed near-miss model (RF-A + sigmoid calibrator)
<code>models/near_miss/{dummy_baseline,logistic_regression_A/B/D,random_forest_A/B/D,gradient_boosting_A/B/D}.joblib</code>	3	Experiment pipelines (10 total)
<code>models/accident/calibrated_accident_detector.pth</code>	6	Deployed accident detector (Phase 5B EXP_A weights + Platt calibrator)
<code>models/accident/phase5b_expA_best.pth</code>	5B	Selected raw checkpoint
<code>models/accident/phase5b_expB_best.pth</code> , <code>phase5b_expB_best_last.pth</code> , <code>phase5b_expA_best_last.pth</code> , <code>phase5b_interrupted_checkpoint.pth</code>	5B	Experiment/interrupted checkpoints (retained for future accuracy work)
<code>models/accident/{best,final}_detector.pth</code> , <code>model_metadata.json</code>	5	Immutable broken baseline (kept for the record)
<code>models/accident/calibration_metadata.json</code>	6	Calibrator params, fit metrics, dataset SHA-256 hashes, leakage guard
<code>models/accident/phase5b_checkpoint_metadata.json</code>	5B	EXP_B interrupted-state metadata

src/

File	Phase	Purpose
src/data_preparation.py	2	Deterministic Phase 2 pipelin (re-runnable)
src/near_miss/{split,train,evaluate,calibrate,probability_pipeline,reporting,plotting}.py	3-4	Near-miss split/train/eval/calibrate + inference pipeline
src/accident/{labels,model,dataset,transforms,train,inference,evaluate,metrics,reporting,visualization,debug_visuals,utils}.py	5-5B	Detector library (labels.py = label-space single source of truth)
src/accident/{run_phase5,run_phase5b,phase5b,phase5b_analysis,phase5b_report}.py	5, 5B	Baseline + improvement experiment drivers
src/accident/{calibration,calibrated_detector,phase6,phase6_report}.py	6	Signal aggregation, Platt calibration, Phase 6 pipeline report
src/config.py	7	All fusion weights/thresholds/paths
src/risk_engine.py	7	Pure fusion (assess_risk , compute_risk_level , recency_factor)
src/safety_intelligence.py	7	Orchestrator (lazy model loading, assess_event / assess_ima

reports/

File	Phase
dataset_inventory.md , near_miss_dataset_audit.md , accident_dataset_audit.md , data_quality_report.md , dataset_audit.md , phase1_summary.json	1
phase2_data_preparation.md , data_dictionary.md , class_balance_phase2.json	2
near_miss/{split_report,model_evaluation,leakage_review}.md , near_miss/{model_comparison,model_comparison_phase4,feature_importance,threshold_analysis,threshold_analysis_calibrated,calibration_comparison}.csv , near_miss/model_results.json , near_miss/phase4_summary.json	3-4
near_miss/{calibration_report,error_analysis,robustness_analysis,probability_contract}.md , near_miss/false_{negatives,positives}_test.csv	4
accident/{environment_report,dataset_summary,validation_results,test_results,error_analysis,accident_detection_contract}.md , accident/phase5_summary.json , accident/training_{config,history}.json	5
accident/phase5b_experiment_log.md , phase5b_experiments.csv , phase5b_exp_{a,b}_history.json , phase5b_exp{A,B}_run.log	5B
accident/{calibration_results,test_results_phase6,phase6_summary,probability_contract}.md , accident/{phase6_summary.json,test_results.json,threshold_analysis.csv} , accident/phase6_run.log	6
safety/phase7_summary.md	7
member4/Member4_Final_Report.md (+ PDF)	8

data/processed/

File	Purpose
near_miss/dtc_fm_combined.csv / .parquet	Combined features+labels, 2,508 rows (Parquet = exact floats)
near_miss/splits/{train,validation,test}.parquet , split_assignments.csv	Reproducible 70/15/15 split
near_miss/{duplicate_report.csv,label_mapping.json,feature_policy.json}	Duplicate groups, neutral label map, per-feature decisions
accident/{train,validation,test}_index.parquet	2,122/316/325 index rows → (source_file, row_index) + SHA-256 + annotations
accident/{duplicate_report.csv,split_integrity_report.json,label_mapping.json,corruption_report.json}	Integrity artifacts (all PASS)

tests/ and figures/

- tests/ — 6 files, 119 tests (§9).
- audit_scripts/ — csv_audit.py , parquet_audit.py , parquet_labels_audit.py , bbox_format_probe.py .
- reports/figures/near_miss/ — 9 PNGs (ROC/PR curves, confusion matrix, threshold, calibration comparison, FN/FP analyses, feature importance).
- reports/figures/accident/ — training curves, calibration curve, confidence-threshold curve, reliability diagram, GT example grids, 24 debug panels, 8 test-example panels.

11. Limitations

Stated as recorded in the project’s own reports — no softening:

1. **Unconfirmed DTC-FM label semantics.** No label map exists in the files or the local documentation. `class_1` is only *statistically consistent* with a conflict/near-miss. Every metric and output uses neutral `class_0/class_1`; interpreting `class_1` as "dangerous" requires authoritative documentation. This blocks final safety-oriented framing of all near-miss results.
2. **PET is post-event/offline.** PET and `FE_inv_PET` are computable only after an interaction completes; the offline near-miss model is **not real-time-deployable as-is**. Removing them (config D) costs ~0.02 ROC-AUC and ~0.06–0.08 PR-AUC but stays usable (ROC ≥ 0.90).
3. **Accident dataset characteristics.** ~99% accident-positive frames (image level) make event-level classification/labeling weakly identifiable; object-level test class mix drifts +8.8 points vs train; 26 train-internal duplicate images; small validation/test sets (316/325 images) → high metric variance; letterboxed 640×640 frames; train-only flip/brightness augmentation by the provider.
4. **Accident model accuracy.** Test accident AP@50 0.6706 / mAP@50 0.4381 / accident recall 0.4409 — usable but modest; 5 of 324 accident-positive test images were missed entirely at event level; the single negative image was a confident event-level FP; accuracy improvement was explicitly deferred, not attempted.
5. **Calibration limitations.** Near-miss calibration fitted on n=376 (small); isotonic rejected as overfit. Accident event-level calibration is prevalence-bound (ECE 0.0176 on test with only two populated reliability bins) and must be re-fit on deployment data with different prevalence.
6. **Risk score is a heuristic.** No ground truth for "risk"; no calibration or validation of the score itself; weights/thresholds are engineering choices, not fitted values; linear fusion chosen for transparency; missing-input renormalization can understate risk.
7. **Severity and recency have no data source.** Neither dataset carries severity labels or timestamps; both are caller-supplied under a documented contract.
8. **No real-world deployment validation.** Everything is validated on this dataset's own splits. No live CCTV stream, no cross-camera transfer test, no operational latency measurements. Camera-dependent pixel features (`target_dist_px` family) may not transfer across camera geometries.
9. **Compute constraints.** CPU-only training (torch CPU, 8 threads) motivated 320×320 inputs, 2-epoch baseline, and the EXP_B interruption; 640×640 training would be ~4× slower per epoch on this machine.
10. **DTC-FM engineering caveats.** Positional alignment only (no IDs); 5 benign duplicate groups; `FE_safety_index` numerically unstable (1e-6 floor); acceleration values quantized (~0.56 m/s² steps); HCV class appears in 2 rows.

12. Member 5 Handoff

12.1 What Member 5 receives

Input	Form	Notes
Near-miss/conflict probability	<code>SafetyIntelligence.assess_event(...)</code> field <code>near_miss_probability</code> — calibrated P(<code>class_1</code>) from <code>random_forest_A_calibrated.joblib</code>	Label semantics unconfirmed; carry the caveat
Accident probability	field <code>accident_probability</code> — calibrated P(image contains accident) from <code>calibrated_accident_detector.pth</code>	Per-image/event, not per-box; prevalence-bound calibration
Per-object detections	<code>detector_output.detections</code> — bbox <code>[x1,y1,x2,y2]</code> , category, uncalibrated confidence	For localization/display, not decisions
Raw detector signal	<code>detector_raw_signal.accident_detection_confidence</code>	Ranking score, explicitly NOT a probability
Fused risk assessment	<code>risk_score</code> (0–100), <code>risk_level</code> (LOW/MEDIUM/HIGH/CRITICAL)	Heuristic, <code>is_probability: false</code> , not validated
Evidence completeness	<code>present_inputs</code> , <code>missing_inputs</code> , <code>weights_applied</code> , <code>contributions</code> , <code>recency_factor</code>	Member 5 can gate on evidence completeness
Transparency	<code>explanation</code> list (formula, inputs, contributions, caveats)	Suitable for display/audit
Configurability	<code>src/config.py</code> (+ <code>risk_config</code> override)	Weights, thresholds, half-life, artifact paths — no code changes needed
Contracts	<code>near_miss/probability_contract.md</code> , <code>accident/probability_contract.md</code> , <code>accident/accident_detection_contract.md</code>	Schema + guarantees + non-guarantees + update policy
Provenance	<code>model_provenance</code> block + <code>calibration_metadata.json</code>	Artifact paths, fit splits, dataset hashes

12.2 What remains Member 5's responsibility

- **All signal-control / decision-making.** Choosing any deployment threshold (Member 4 deliberately fixed none; the validation threshold tables are provided for that choice), mapping risk levels to actions, actuation, or intervention logic.
- **Supplying severity and event_age_hours** (or deciding not to) under the documented [0, 1] / hours contracts — Member 4 does not measure either.
- **Deciding how to treat missing_inputs** (gate, alert, or abstain) — Member 4 reports them but does not decide.
- **Event aggregation, streaming, latency, UI, and operational integration** — Member 4's layer is per-event/per-image batch inference.
- **Any production recalibration** if the deployment distribution differs (per the update policies in the probability contracts: refit on a fresh validation-like split, never on TEST; bump `detector_version`).

12.3 Explicit boundary

Per the Phase 7 rules and the risk engine's own output text: the risk score "must not be used to control traffic signals." Member 4 produces **assessments only**; no signal-control, actuation, or intervention path exists in Member 4's code, and none should be attributed to it.

13. Future Work (intentionally deferred)

Accuracy improvement (explicitly deferred by Phase 6 rules): - Complete the interrupted EXP_B 25-epoch run (checkpoint retained) and add EXP_D-class changes: longer schedules, 640×640 resolution, augmentation, LR/scheduler tuning, backbone comparison from scratch. - Address the two event-level failure modes: missed accident-positive images (5/324) and the confident event-level FP on the single negative image. - Larger/more diverse accident data; per-class metrics are limited by the 316/325-image eval splits.

Near-miss system: - Replace post-event PET with pre-event surrogates (projected time-to-collision, distance rate) for a real-time-deployable variant (config D is the documented baseline). - Confirm DTC-FM label semantics from the dataset provider and re-frame all class_0/class_1 metrics in operational terms. - Retire or stabilize `FE_safety_index`-style composites; camera calibration or metric-only features for cross-camera transfer.

Calibration and validation: - Re-fit the accident calibrator on deployment-prevalence data; collect a real-world validation set for both models. - Validate or empirically derive the risk-fusion weights (currently engineering choices); consider grounding severity in a defined scale.

Engineering: - GPU training (the entire 5B/6 pipeline was CPU-bound); batch/streaming inference service; persistence of assessments; monitoring for drift.

14. Final Summary

What has Member 4 successfully built by the end of Phase 7?

A complete, tested, and documented safety-intelligence layer that turns two raw datasets into one fused, explainable risk assessment:

1. **Audited and prepared both datasets** without modifying a single original file — 2,508 DTC-FM interaction records and 2,763 accident CCTV frames (5,400 boxes), with quality grades (GOOD), zero cross-split contamination (SHA-256-verified), resolved engineered-feature formulas, documented label-mapping status, and reproducible, leakage-safe splits.
2. **Built and calibrated a near-miss/conflict probability model** — 10 trained pipelines, a validation-only selection of Random Forest (config A), and sigmoid calibration: test ROC-AUC 0.929, PR-AUC 0.830, calibrated test Brier 0.0898 / ECE 0.0439 — with the honest caveat that PET makes the primary configuration offline-only and the label semantics remain unconfirmed.
3. **Built and calibrated an accident detection system** — diagnosing and fixing a confirmed background/label-collision bug that had driven the Phase 5 baseline's accident AP to 0.0, then training Faster R-CNN to validation accident AP@50 0.7624, selecting the checkpoint on validation only, calibrating the `noisy_or_top5` event signal with Platt, and reporting a single frozen test evaluation: accident AP@50 0.6706, precision 0.6949 / recall 0.6292 @0.5, event-level Brier 0.0169.
4. **Fused both probabilities with severity and recency** into a transparent, configurable, and fully tested risk score (weights 0.35/0.40/0.15/0.10; thresholds 25/50/75; 24-hour recency half-life) that renormalizes over present evidence, refuses to guess on invalid or missing inputs, never treats raw detector confidence as a probability, and explains every assessment it emits.
5. **Delivered the Member 5 interface:** `SafetyIntelligence.assess_event / assess_image` returning the JSON payload shown in §7 — verified here on real data (risk 60.60, HIGH) — plus written probability contracts, update policies, and an explicit boundary: Member 4 assesses risk; it does not control signals, measure severity, or decide actions.

Everything is backed by 119 passing tests, per-phase reports, and honest limitation registers. The system is research-grade, not deployment-grade: the accuracy, calibration-prevalence, label-semantics, and real-world-validation gaps are documented in §11 and queued in §13 for the phases that follow.

Report generated 2026-09-03 from the project's own artifacts. All metrics were cross-checked against `reports/`, `models/*/...metadata.json`, and a fresh full test run (119/119 passed). No dataset, model, or existing report was modified to produce this document.